



Amelioration of postpartum hyperketonemia using amino acids, cyanocobalamin, inositol, α -lipoic acid, or monensin during the transition period of dairy cows

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ABSTRACT

Strategies for the prevention of hyperketonemia, or subclinical ketosis, based on providing additional glucose or glucose precursors provide discordant results suggesting that other poorly understood metabolic factors may potentially contribute to the onset of this affliction. The aim of this study was to compare the efficacy of hepatic-protectors (acetyl-Met, Ala, Arg, Thr, Gln, cyanocobalamin, inositol, and α -lipoic acid) or monensin to reduce postpartum hyperketonemia. A total of 594 Holstein Friesian dairy cows from a single herd were enrolled in a longitudinal study. Animals were blocked by parity and randomly distributed into 4 groups at dry-off: control group (CTR; $n = 148$); complete preventive protocol based on hepato-protectors (CPP; $n = 149$); simplified preventive protocol based on hepato-protectors (SPP; $n = 149$); and monensin (MON; $n = 148$). Cows in CPP received an i.m. injection of 2 mL/10 kg of BW of a solution containing acetyl-Met, Ala, Arg, Thr, Gln, inositol, and cyanocobalamin at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL/animal of a solution containing acetyl-Met, α -lipoic acid, and cyanocobalamin at calving, and 2, 4, 6, 8, 10, and 12 d postpartum. The SPP group received both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively) at 12 d before the expected calving and 6 d postpartum. The MON group received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date. Blood samples were collected at 21 and 7 d before expected calving, the day of calving, and 7, 14, 28, and 55 d postpartum. Blood nonesterified fatty acids (NEFA) and BHB were determined in all animals, whereas blood biochemical profiles were evaluated in 180 cows

(30.3%) randomly selected. Cows in CPP and SPP had lower serum NEFA, BHB, aspartate amino-transferase, γ -glutamyl transferase, and urea concentrations than CTR and MON cows. Furthermore, SPP cows had the greatest serum albumin concentrations during the postpartum period. Postpartum hyperketonemia prevalence was lower in CPP and SPP than in CTR cows at 14 and 28 d, and lower than MON cows at 55 d. In addition, hyperketonemia prevalence was lower in SPP than in CTR cows at 55 d postpartum. Hepato-protectors influenced energy and protein metabolism, and liver metabolism of cows, and they may represent a valuable strategy to reduce hyperketonemia in dairy cattle.

Key words: biochemistry, ketosis, metabolism, prevention, transition

INTRODUCTION

The transition period, defined as the 3 wk before and 3 wk after calving, is considered a critical phase for dairy cows. Increased energy requirements for fetal growth and colostrum and milk production are associated with increased demands of glucose, AA, and calcium, among other nutrients. Cows typically experience a negative energy balance (NEB) after calving (Fiore et al., 2015; Horst et al., 2021). This NEB is characterized by body resources mobilization, from both adipose and muscular tissue, altered oxidative state characterized by increased oxygen consumption and fatty acid oxidation, and altered inflammatory and immune responses. If nonesterified fatty acids (NEFA) mobilized from adipose tissue are not fully metabolized, then animals may develop metabolic stress with an increased risk of metabolic diseases (Abuelo et al., 2015, 2019; Coleman et al., 2020). Excessive NEFA mobilization may lead to serum concentrations above 0.57 mEq/L in the postpartum and part of these NEFA may result in ketogenesis (partial oxidation of fatty acids) with an accumulation of ketone bodies in

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

the form of acetoacetate, acetone, and BHB (Ospina et al., 2010; Tessari et al., 2021). The oxidation of NEFA generates large amounts of acetyl-CoA, progressively reducing the intermediate metabolites of the tricarboxylic acid (TCA) cycle (Horst et al., 2021). Some AA, such as Met, Ala, Arg, Thr, Gln, can be used to produce the cycle's intermediate metabolites. Consequently, a shortage of these AA can compromise the TCA cycle causing a potential block in one or more steps leading to ketogenesis (Lisuzzo et al., 2022b; Fiore et al., 2023).

Among ketone bodies, BHB is the most stable, and it is used for the diagnosis of subclinical ketosis with a proposed threshold serum concentration of 1.0 to 1.2 mmol/L of blood (Ospina et al., 2010; Mann and McArt, 2023). Subclinical ketosis or hyperketonemia represents one of the most important metabolic disorder in dairy cows that usually occurs within the first 2 to 7 wk after calving (Fiore et al., 2014; Mann and McArt, 2023; Rico and Barrientos-Blanco, 2024). Previous studies (Rodríguez et al., 2017; Benedet et al., 2019; Loiklung et al., 2022) report prevalence and incidence of ketosis ranging between 11% and 47%, and 20% and 44%, respectively. Hyperketonemia has been linked to increased risk of metabolic, reproductive, and infectious disorders (Horst et al., 2021). Usually, this disease represents a risk factor during the first 2 wk after calving. Whereas the effects on milk production and reproduction are sometimes conflicting (Hubner et al., 2022; Rodriguez et al., 2022).

A common preventive strategy for hyperketonemia consists of supplementing monensin in the ration. Monensin is an ionophore that reduces gram-positive and promotes gram-negative bacteria in the rumen (Mammi et al., 2021) with a consequent increase in the production of propionate, a glucose precursor, in the rumen. However, the extensive use of an antimicrobial such as monensin in livestock raises concerns about the development of antimicrobial resistance in nontarget microorganisms in the intestinal microbiota (Carresi et al., 2024). In addition, correction and prevention of hyperketonemia through glucogenic precursors has provided inconsistent results suggesting that other poorly understood metabolic factors may contribute to the onset of hyperketonemia (Rico and Barrientos-Blanco, 2024). Alternative protocols using niacin, an antilipolytic agent, and anti-inflammatory drugs were evaluated with positive results (Chirivi et al., 2024). Another alternative preventive protocol for hyperketonemia is based on supporting the liver metabolism through AA precursors of the TCA cycle, vitamins as cyanocobalamin, inositol (insulin mimetic effect), and α -lipoic acid (anti-oxidant action; Goręca et al., 2011; Croze and Soulage, 2013; Fiore et al., 2016; Fiore et al., 2021). The hypothesis herein was that preventive protocols that support liver metabolism may provide different metabolic response

compared with other strategies based on providing glucogenic precursors. The aim of this study was to compare the efficacy and consequences of preventive protocols for hyperketonemia based on monensin and hepatic-protectors containing glucogenic AA (acetyl-Met, Ala, Arg, Thr, Gln), cyanocobalamin, inositol, and α -lipoic acid during the transition period of dairy cows.

MATERIALS AND METHODS

Animals

Animal care and procedures were conducted in accordance with the Guide for the Care and Use of Laboratory Animals and Directive 2010/63/EU for animal experiments (National law: D.L. 26/2014). The ethics statement was approved by the Animal Care and Use Committee of the University of Padua (protocol number 204359/2023). The study was conducted in a freestall barn located in Friuli-Venezia Giulia (Italy). A total of 594 cows were enrolled in this study (145 during winter, 146 during spring, 162 during summer, and 141 during fall). Primiparous cows (**PPC**; $n = 166$) and multiparous cows (**MPC**; $n = 428$) were kept in separate pens. The dry period began about 60 d from the expected calving with a steaming-up during the last 3 wk of pregnancy. The diet of the dry period is depicted in Supplemental Table S1 (see Notes). During the prepartum, cows received anionic salts in the diet to induce a dietary negative cation-anion difference. Urine pH was monitored weekly for assessing animal response to the anionic diet, with a target urine pH of 6.1. After calving, cows were fed a TMR twice daily and milked 3 times per day (0600, 1600, and 2200 h) and individual milk production recorded. Clinical examination was performed weekly by the farm veterinarian to identify postpartum diseases including retained placenta, metritis, abomasal displacement, milk fever, and ketosis. All animals were artificially inseminated (**AI**) between 80 to 90 DIM following a synchronization protocol. Cows were checked for pregnancy via ultrasound at 32 and 68 d after insemination. Cows confirmed pregnant at 32, but declared open at 68 d were considered as having a pregnancy loss. Water was provided ad libitum for all animals.

Treatments

A longitudinal study design was applied in the study. Animals were blocked by parity and randomly distributed into 4 groups at the beginning of the dry period: (1) control (**CTR**; $n = 148$); (2) complete preventive protocol (**CPP**; $n = 149$); (3) simplified preventive protocol (**SPP**; $n = 149$); and (4) monensin (**MON**; $n = 148$). The person responsible for group assignment and blocking was not involved in farm and animal management, sampling, and treatment

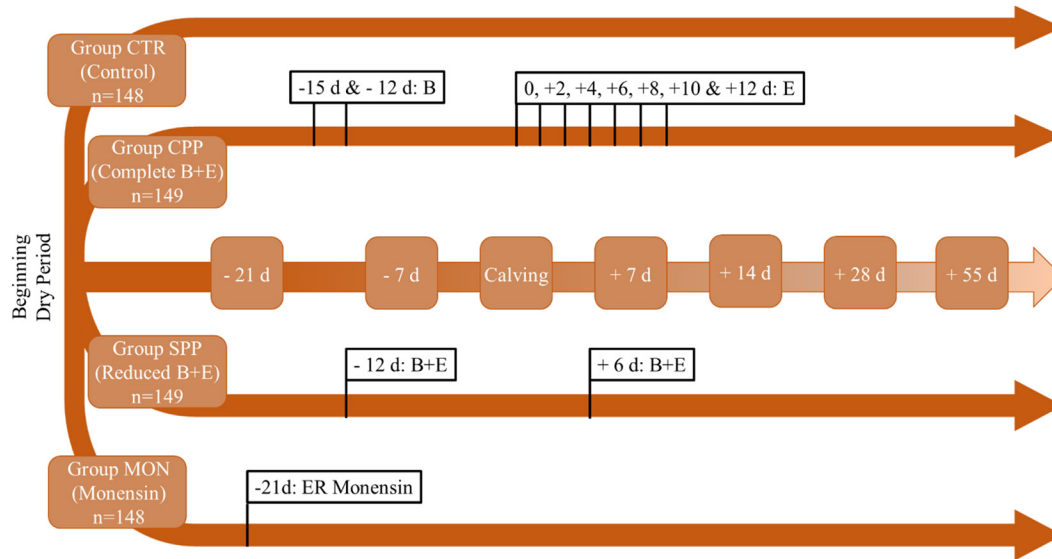


Figure 1. Flowchart of the study design. All lactating animals of the herd ($n = 594$) were enrolled and equally divided in the 4 groups at the beginning of the dry period. Subsequently, a total of 180 animals were equally selected by each group for a complete biochemical profile. The central axis indicated the time points with animals' clinical evaluation and blood sampling. B indicates the first hepato-protector containing AA (acetyl-Met, Ala, Arg, Thr, Gln), inositol, and cyanocobalamin; E indicates the second hepato-protector containing acetyl-Met, α -lipoic acid, and cyanocobalamin; and ER indicates endorumenal bolus of monensin.

administration. Cows in the CPP and SPP groups received the same combination of 2 hepato-protectors, but differed in the application protocol and dose. One hepato-protector contained AA (150 mg/mL acetyl-Met, 15 mg/mL Ala, 15 mg/mL Arg, 15 mg/mL Thr, 15 mg/mL Gln), 30 mg/mL inositol, and 1 mg/mL cyanocobalamin (Bograss, Ceva Salute Animale, Milan, Italy), and the second hepato-protector contained 4.8 mg/mL acetyl-Met, 0.25 mg/mL α -lipoic acid, and 0.5 mg/mL cyanocobalamin (Erbacolina Plus, Ceva Salute Animale, Milan, Italy). Cows in CPP received an i.m. injection of 2 mL of Bograss/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus at calving, and 2, 4, 6, 8, 10, and 12 d postpartum. The SPP group received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively). Cows in the MON group received one intraruminal bolus of monensin (32.4 g/animal, Kexxstone, Elanco Italia, Milan, Italy) 21 d before the expected calving date. A pictorial representation of treatments is depicted in Figure 1. The administration of CPP and MON was performed following the drug guidelines. Instead, the prepartum administration in SPP group was selected among the prepartum administration of the CPP to be the nearest to calving; whereas the postpartum administration in SPP group was selected as a mean value of the postpartum administration in CPP group.

Clinical examinations and blood samplings were performed about 4 to 5 h after the TMR distribution at -21, -7, 0, 7, 14, 28, and 55 d relative to calving. Blood sam-

ples were obtained from the coccygeal vessel using evacuated tubes and stored in Venosafe tubes (9 mL; Terumo Venosafe, Leuvel, Belgium) containing a clot activator. Thereafter, tubes were centrifuged (Heraeus Labofuge 400, Thermo Scientific, Milan, Italy) at $1,750 \times g$ for 10 min at room temperature (around 20°C). The obtained sera for each cow were equally divided in four 1.5-mL Eppendorf tubes. The aliquots were then refrigerated at 4°C and transported in a portable freezer (CoolFreeze CFX65 W professional, Dometic, Stockholm, Sweden; minimum temperature -22°C) at the same constant temperature to the laboratory of the Department of Animal Medicine, Production and Health (MAPS) of the University of Padua (Legnaro, PD, Italy) within 3 h, where they were stored at -20°C until biochemical analysis.

Biochemical Analyses

All biochemical analyses were conducted using an automatic clinical chemistry analyzer (BT3500 Biotecnica Instruments, Rome, Italy). All animals ($n = 594$) were tested for serum BHB concentration by RANBUT RX Monza test (Randox, Crumlin, UK) and NEFA concentration by a colorimetric method (NEFA RX Monza test; Randox, Crumlin, UK). A subsample of 180 animals (30.3% of the herd) equally distributed in the 4 groups and composed by two-thirds MPC (30 of 45 animals per group) and one-third of PPC (15 of 45 animals per group) were randomly selected to determine serum concentration of glucose, total protein, albumin, urea,

creatinine, total bilirubin, alanine amino-transferase (ALT), aspartate amino-transferase (AST), γ -glutamyl-transferase (GGT), lactate dehydrogenase (LDH), creatinine kinase (CK), total cholesterol, and triacylglycerol. The selection was performed to exclude animals affected by postpartum diseases, maintaining equal numbers between groups and an equal ratio between PPC and MPC. Serum concentrations of cholesterol, bilirubin, and albumin at 7 and 28 d postpartum were used to calculate the liver functionality index (LFI) following Trevisi et al. (2011). Specifically, the subindex for albumin (g/L) and cholesterol (mmol/L) was calculated, respectively, as the sum of 50% of the value at 7 d, and 50% of the difference between 28 and 7 d. The bilirubin (μ mol/L) subindex was calculated as the sum of 67% of the value at 7 d, and 33% of the difference between 7 and 28 d. Subsequently, the subindexes were used to calculate the LFI using the formula proposed by Trevisi et al. (2011): $LFI = (\text{albumin} - 17.71)/1.08 + (\text{cholesterol} - 2.57)/0.43 - (\text{bilirubin} - 6.08)/2.17$.

Statistical Analysis

The minimum sample size was calculated a priori using G*Power software v.3.1.9.7 (<http://www.gpower.hhu.de/>). A minimum sample size was 23 animals was obtained considering blood concentration of BHB as a primary outcome among preventive protocols for hyperketonemia presented by Fiore et al. (2021) using an α of 0.05, an β of 0.05 (95% power), a SD of 0.06 mmol/L and an effect size of 0.4 mmol/L. For data related to reproductive performance, sample size was determined using a power analysis conducted with JMP (version 18.1, SAS Institute Inc., Cary, NC) aiming to detect a difference in conception rate of 16 percentage units with a power of 0.70 and an α of 0.05, which yielded a minimum of 135 cows.

All parameters were normally distributed as assessed by Shapiro–Wilk test. Serum NEFA and BHB concentrations were analyzed using a mixed-effects model for repeated measures accounting for the fixed effects of treatment (CTR, CPP, SPP, and MON), sampling time, and their interaction, plus the random effect of cow nested within treatment. Blood biochemical profiles of selected animals ($n = 180$) were also analyzed using a mixed-effects model for repeated measures accounting for the fixed effects as treatment (CTR, CPP, SPP, and MON), sampling time, parity (PPC and MPC), season (winter, spring, summer, and fall), and their interactions, plus the random effect of cow nested within treatment. Milk yield data were analyzed using a mixed-effects model accounting for the fixed effects of preventive protocol, day of lactation, and their interaction, plus the random effect of cow, with day of lactation entering the model as a repeated measure with an autoregressive structure. Post hoc pairwise comparisons

among treatments were performed using Tukey correction for the 3 models. The effect of preventive treatments on pregnancies/AI, pregnancy losses, and proportion of cows pregnant was assessed using a logistic regression model with treatment as a fixed effect.

Prevalence of hyperketonemia ($BHB \geq 1.2$ mmol/L) during postpartum period was analyzed with a repeated mixed-effects model (PROC GLIMMIX) and a binomial distribution. Treatment, time, parity, season, and their interactions were used as fixed effect, whereas cow nested within treatment was used as random and repeated effect.

Lastly, a logistic regression analysis was performed using 2 categories based on blood BHB values (0: no hyperketonemia with $BHB < 1.2$ mmol/L; and 1: hyperketonemic animal with $BHB \geq 1.2$ mmol/L). The model considered the risk factors of treatment, time, parity, and season, and the biochemical parameters selected by a stepwise linear regression (NEFA, glucose, LDH, ALP, total bilirubin, and AST).

P -values ≤ 0.05 were considered significant, whereas values between 0.05 and 0.10 were considered near significance.

RESULTS AND DISCUSSION

General Characteristics and Hyperketonemia Prevalence

Average sampling day before actual calving for the target of 21 d was 27 ± 6.7 d, and that for the target of 7 d preceding calving was 9 ± 5.2 d. Serum NEFA ($P < 0.001$) and BHB ($P = 0.007$) concentrations differed among groups over time (Table 1). Serum NEFA concentrations increased from pre- to postpartum with a peak between 14 and 28 d. At the same time points, both CTR and MON cows had serum NEFA concentrations indicative of high lipid mobilization (≥ 0.57 mEq/L). Serum BHB peaked between 7 and 28 d with no cows in any of the treatment having values indicative of hyperketonemia. Nevertheless, CPP and SPP cows had the lowest serum BHB concentrations at the same time points.

Elevated serum BHB concentrations are also indicative of metabolic stress (Abuelo et al., 2015, 2019). Furthermore, CTR and MON cows experienced an excessive lipomobilization as indicated by serum NEFA ≥ 0.57 mEq/L at 14 and 28 d after calving. On the contrary, serum BHB level was not found above the cut-off threshold of hyperketonemia ($BHB \geq 1.0$ mmol/L) proposed by Ospina et al. (2010). Both, PPC and MPC in the CTR group had serum BHB concentrations indicative of hyperketonemia at d 28 and 14 of lactation, respectively. The different time relative to calving when serum BHB concentration peaked according to parity was probably due to differences in milk yield.

Prevalence of hyperketonemia for the whole herd is depicted in Table 2. Prevalence of hyperketonemia

Table 1. Serum concentration of nonesterified fatty acids (NEFA) and β -hydroxybutyrate (BHB) between 21 before and 55 d after calving as influenced by treatments (n = 594)

Item	Treatment ¹				SEM	P-value		
	CTR (n = 148)	CPP (n = 149)	SPP (n = 149)	MON (n = 148)		Treatment	Time	T × t ²
NEFA (mEq/L)								
Days relative to calving								
–21 d	0.11 ^c	0.12 ^c	0.11 ^c	0.13 ^c	0.02	0.888	<0.001	<0.001
–7 d	0.15 ^c	0.15 ^c	0.13 ^c	0.14 ^c				
0 d	0.37 ^d	0.41 ^b	0.39 ^b	0.39 ^d				
7 d	0.47 ^{c,x}	0.41 ^{b,y}	0.41 ^{b,y}	0.45 ^{c,xy}				
14 d	0.59 ^{b,x}	0.49 ^{a,y}	0.47 ^{ab,y}	0.57 ^{b,x}				
28 d	0.69 ^{a,x}	0.51 ^{a,y}	0.51 ^{a,y}	0.67 ^{a,x}				
55 d	0.46 ^{c,x}	0.41 ^{b,xy}	0.40 ^{b,y}	0.45 ^{c,x}				
BHB (mmol/L)								
Days relative to calving								
–21 d	0.65 ^c	0.66 ^b	0.65 ^b	0.66 ^{cd}	0.03	0.997	<0.001	0.007
–7 d	0.63 ^c	0.63 ^b	0.62 ^b	0.63 ^d				
0 d	0.59 ^c	0.58 ^b	0.59 ^b	0.57 ^d				
7 d	0.83 ^{b,x}	0.72 ^{a,y}	0.75 ^{a,y}	0.73 ^{c,y}				
14 d	0.91 ^{a,x}	0.70 ^{a,y}	0.71 ^{a,y}	0.88 ^{a,x}				
28 d	0.88 ^{ab,x}	0.71 ^{a,y}	0.74 ^{a,y}	0.87 ^{a,x}				
55 d	0.66 ^{c,y}	0.63 ^{b,y}	0.60 ^{b,y}	0.82 ^{b,x}				

^{a-c}Values with different letters differ at $P < 0.05$ over time within treatment.

^{x,y}Values with different letters differ at $P < 0.05$ between treatments within time.

¹CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss (Ceva Salute Animale, Milan, Italy)/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus (Ceva Salute Animale) at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date.

²T $\times$ t = effect of the interaction between treatment and time.

ranged between 1.04% and 30.4% and it was generally lower in CPP and SPP than in the CTR and MON groups. Specifically, the hyperketonemia prevalence of CPP and SPP cows were lower than MON cows at 14, 28, and 55 d, whereas it was lower than CTR cows at 28 and 55 d. Moreover, the hyperketonemia prevalence was generally greater during spring than fall. Prevalence of hyperketonemia herein was at the low end of values reported in the literature ranging between 11.2% and 47.2% (Benedet et al., 2019; Cainzos et al., 2022). The use of hepato-protectors (CPP and SPP) generally reduced the prevalence of about 12% compared with CTR, and about 18% compared with MON. The reduction was between 14% and 17% compared with MON at 14 d; about 21% compared with CTR and 24% compared with MON at 28 d; and about 7% compared with CTR and 22% compared with MON at 55 d.

Complete Biochemical Analyses

Regarding serum biochemical profiles (Figure 2) measured in randomly selected cows (n = 180), 9 parameters differed among groups: NEFA ($P = 0.012$), BHB ($P = 0.023$), AST ($P = 0.001$), GGT ($P = 0.010$), LDH ($P = 0.011$), albumin ($P = 0.027$), total protein ($P < 0.001$), and urea ($P = 0.002$). However, differences in serum

glucose ($P = 0.056$) and CK ($P = 0.079$) concentrations did not reach significance among treatments. However, these parameters were affected by an interaction between treatment and parity. Similarly, an interaction was found for LFI ($P = 0.037$) among treatment, and parity (Figure 3). No interactions were observed between treatment and season for any of the parameters evaluated. Serum concentrations for all cows are shown in Supplemental Table S2 (see Notes), whereas serum concentrations for PPC are depicted in Supplemental Table S3 (see Notes), and those of MPC in Supplemental Table S4 (see Notes).

Glucose Metabolism

All MPC had a general hypoglycemic state probably related to a greater milk yield compared with PPC at 14 and 28 d along with a high lipomobilization at 14 d. However, only MPC within the CTR group had an hyperketonemic state at the same time point. Serum glucose concentrations peaked at calving in all groups with the highest value in the SPP group, especially in MPC, and the lowest in the CTR group. Furthermore, CTR cows continued to have lower serum glucose concentrations also at 7 d after calving, which would indicate a general hypoglycemic state. Primiparous cows within the CTR group experienced an increased lipomobilization concurrent

Table 2. Prevalence and incidence of hyperketonemia (serum BHB > 1.2 mmol/L) at 7, 14, 28, and 55 d postpartum as affected by treatment¹ (n = 594)

	General effect (%)	7 d (%)	14 d (%)	28 d (%)	55 d (%)	
Treatment	Prevalence Incidence	Prevalence Incidence	Prevalence Incidence	Prevalence Incidence	Prevalence Incidence	<i>P</i> -value ²
CTR	17.3 ^x	13.4 ^b	22.3 ^{a,xy}	23.7 ^{a,x}	8.9 ^{b,y}	Treatment = 0.011 Time = 0.004 T × t = 0.013
	37.8	13.5	10.9	12.3	8.0	
CPP	5.34 ^y	10.3 ^b	16.7 ^{a,y}	2.26 ^{c,y}	1.04 ^{c,z}	
	20.8	10.1	8.21	2.44	1.67	
SPP	5.48 ^y	9.23 ^{ab}	13.3 ^{a,y}	2.37 ^{b,y}	1.57 ^{b,z}	
	16.8	9.40	4.44	2.33	1.59	
MON	23.9 ^x	14.4 ^b	30.4 ^{a,x}	26.4 ^{a,x}	23.3 ^{a,x}	
	50.0	14.2	19.7	13.7	15.9	
General effect		11.7 ^b	20.8 ^a	13.7 ^b	8.62 ^b	

Treatment	PPC (%)	MPC (%)	Winter (%)	Spring (%)	Summer (%)	Fall (%)	<i>P</i> -value ²
CTR	17.0	17.7	21.2	24.5	14.0	9.81	Season = 0.004
CPP	3.60	7.86	4.92	7.79	5.41	1.40	
SPP	7.65	3.90	1.56	14.5	9.09	3.27	
MON	27.1	21.0	26.5	26.9	16.6	23.2	
General effect	11.0	10.6	12.0 ^{ab}	16.9 ^a	11.0 ^{ab}	5.92 ^b	

^{a-c}Values with uncommon superscripts within treatment differ.

^{x-z}Values with uncommon superscripts within day differ.

¹CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date (n = 148).

²*P*-values of parity, treatment × parity, treatment × parity × time, treatment × season, and treatment × season × time were not significant (*P* > 0.05).

with a hypoglycemic state that remained until 28 d after calving despite the return to euglycemic conditions at 14 d. Interestingly, this group exhibited hyperketonemia at 28 DIM along with a continued greater lipomobilization. In contrast, MPC in the MON group did not experienced hypoglycemia, but had an elevated lipomobilization at 7 and 14 d after calving with increased serum BHB concentrations. However, this increase in serum NEFA and BHB concentrations did not lead to hyperketonemia. Instead, PPC in the hepato-protector groups (CPP and SPP) had a low lipomobilization with a progressive decrease in BHB concentration during postpartum.

Herein, hyperketonemia was not always accompanied by hypoglycemia, even if this concurrent status has been defined as an historical characteristic of subclinical ketosis (Lisuzzo et al., 2022a; Rico and Barrientos-Blanco, 2024). Furthermore, serum NEFA and BHB concentrations did not appear to be strongly related to each other considering that the onset of hyperketonemia was not always related to higher lipomobilization (NEFA ≥ 0.57 mEq/L). These findings further evidence that hyperketonemia is not necessarily related to excessive lipomobilization but also to other aspects of animal metabolism.

Liver Health Status

Both serum AST and LDH concentrations peaked at 7 d after calving in all groups, probably related to the energy

metabolism discussed above. All preventive protocols (CPP, SPP, and MON) resulted in lower serum AST concentrations 7 d before calving compared with the CTR group. Similarly, serum GGT concentrations were lower in the hepato-protector groups (CPP and SPP) compared with the CTR and MON groups after calving. The reduction in serum GGT and AST could be a consequence of a lower lipomobilization and metabolic stress (lower serum BHB) in cows treated with hepato-protectors (CPP and SPP). Serum CK concentrations increased above the threshold value of 250 U/L (Cozzi et al., 2011) indicative of muscular catabolism at calving in all groups. This result agrees with the subsequent increase in both serum AST and LDH concentrations. Creatinine kinase is one the markers that raises in blood shortly after muscle catabolism (Kaneko et al., 2008), but it also has the shortest half-life (Kaneko et al., 2008). In contrast, AST increases later than CK and it has a much longer half-life (Kaneko et al., 2008). Therefore, the increase in muscle catabolism following calving may have contributed to the increase in serum AST and LDH observed herein.

Protein Metabolism

Total protein serum concentrations were below the physiological reference range (<67.4 g/L; Kaneko et al., 2008) in all groups around calving. This reduction was also pronounced in the CPP and MON groups 7 d

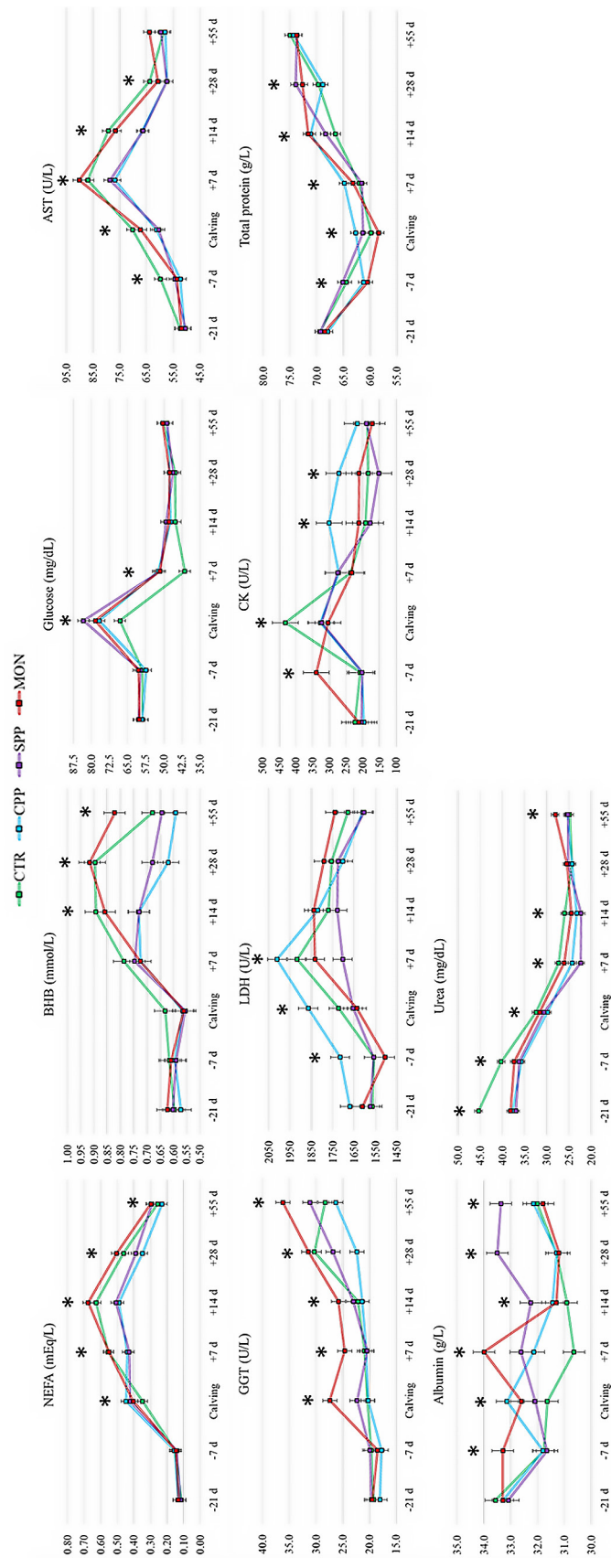


Figure 2. Selected serum analytes in dairy cows ($n = 180$) between 21 d before and 55 d relative to calving as influenced by treatments (CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss (Ceva Salute Animale, Milan, Italy)/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus (Ceva Salute Animale) at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date). *Indicates significant differences among groups; means were presented specifically in Supplemental Table S2 (see Notes). Error bars are SE.

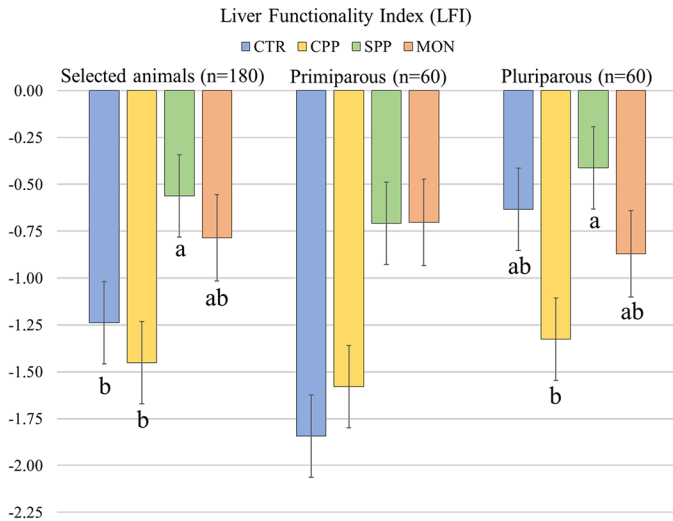


Figure 3. Liver functionality index (based on serum concentration of total cholesterol, total bilirubin, and albumin) at 7 and 28 d postpartum in dairy cows (treatment effect: $P < 0.001$; treatment $\times$ parity effect: $P = 0.037$). CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date. (a,b) Values with uncommon letters within animal category (selected, primiparous, and pluriparous) differ. Error bars are SE.

before calving, but CTR and SPP cows had the lowest values at 7 and 14 d after calving, and CTR and CPP at 28 d after calving. Serum total proteins are composed by several fractions: albumin and globulins (α , β , and γ). Although total serum protein was below the physiological values, this was not the case for albumin (albumin >30.3 g/L) suggesting that the reduction in total protein was mainly driven by changes in the fraction of globulins (Piccione et al., 2011).

Cows in the CTR group had the lowest serum albumin concentration from 7 d before calving until the end of the study, whereas serum CK increased at calving and serum urea concentrations were the highest from -21 to 14 d relative to calving. These results suggest that CTR cows potentially presented a greater peripartum inflammatory condition than the other groups, as well as a higher protein catabolism leading to a consumption of albumin, and a subsequent increase in urea production. Serum albumin concentration at 7 d prepartum decreased in CPP cows followed by an increase at calving followed by a decrease during postpartum. The same group had greater serum CK concentrations, which were above the threshold value of 250 U/L (Cozzi et al., 2011) between calving and 28 d thereafter. Cows in CPP received the highest amount of AA (acetyl-Met, Ala, Arg, Thr, Gln) at 15 and 12 d prepartum, which probably supported this

deficiency until calving. Circulating AA are important substrates for the adequate functioning of the TCA and urea cycles (Coleman et al., 2020; Lisuzzo et al., 2022c). Furthermore, an increased protein metabolism led to a rise in serum urea concentrations (De Koster and Opsomer, 2013; Sadri et al., 2023). Thereafter, the results indicate an increased mobilization of muscle tissue after calving probably to meet AA requirements of the cows. However, lower serum urea concentrations in hepato-protectors' groups (CPP and SPP) could be due to the administration of an anti-oxidant (α -lipoic acid) and cyanocobalamin during the postpartum, which may have ameliorated the inflammatory state and partially supported the TCA cycle functioning, respectively (Goręca et al., 2011; González-Montaña et al., 2020). The MON cows maintained serum albumin concentration up to 7 d postpartum with a subsequent decrease until the end of the study. Serum CK concentrations were above the threshold value for muscular damage at 7 d before calving and at parturition, but it decreased thereafter. In contrast, serum urea concentration in MON was greater than in the SPP cows between calving to the end of the study. Furthermore, the MON cows had an increased protein catabolism (reduced albumin level) and higher urea production during postpartum suggesting that supporting only glucose metabolism did not overcome the need for AA, forcing the cow to use other sources of N such as albumin. Lastly, serum albumin levels increased 7 d after calving in SPP cows having the highest levels compared with all other groups. At the same time, higher serum CK concentrations were evidenced at calving and at 7 d postpartum in SPP, which decreased thereafter. In contrast, serum urea levels were the lowest after 7 d postpartum. In addition, this group received the highest amount of AA at 12 d prepartum and 6 d postpartum suggesting supplementation of acetyl-Met, Ala, Arg, Thr, and Gln also during postpartum reduced protein catabolism and a subsequent urea production.

The LFI and Risk of Hyperketonemia

Serum concentrations of albumin, total cholesterol and total bilirubin were used to measure LFI (Trevisi et al., 2011). Among these parameters, only albumin differed among groups over time and, therefore, probably it was the most important metabolite that influenced the LFI. This index was greater in SPP than in CPP and CTR cows indicating a better liver functioning probably related to the lower protein catabolism and urea production discussed above.

Chances of incurring hyperketonemia were influenced by a triple interaction between treatment ($P = 0.011$), time ($P = 0.013$), and season ($P = 0.003$). The CPP (OR 0.65; 95% CI: 0.20 to 0.91) and SPP (OR 0.43; 95% CI: 0.31 to 0.95) groups showed a protective effect from

Table 3. Reproductive performance of dairy cows after first artificial insemination (AI) as affected by treatments

Parameter	CTR ¹	CPP ¹	SPP ¹	MON ¹	P-value
Pregnancy at 32 d after AI, ² %	49.0 ^b	41.2 ^c	56.1 ^a	50.4 ^b	0.05
OR ³		0.82 (0.40–0.95)	1.12 (1.03–1.79)	1.07 (0.86–1.15)	
Pregnancy at 68 d after AI, ² %	45.4 ^b	36.8 ^b	52.6 ^a	44.5 ^b	0.04
OR ³		0.81 (0.40–1.06)	1.16 (1.05–1.90)	0.98 (0.54–1.45)	
Pregnancy loss, %	8.9	10.8	5.6	10.9	0.09
OR ³		1.21 (0.38–4.04)	0.63 (0.16–2.22)	1.22 (0.41–4.06)	

^{a-c}Values with uncommon superscripts differ at $P < 0.05$.

¹CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs i.m. (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date.

²Pregnancy per AI diagnosed at 32 and 68 d after insemination.

³Odds ratio and 95% CI (in parentheses) using CTR as a reference.

hyperketonemia compared with CTR group ($P = 0.007$ and $P = 0.005$, respectively). Both CPP and SPP showed also a protective effect from hyperketonemia compared with MON group (CPP OR 0.27 and 95% CI: 0.09 to 0.86, $P = 0.026$; SPP OR 0.44 and 95% CI: 0.14 to 0.84, $P = 0.017$). Spring was the season with the greatest risk for hyperketonemia cases (OR fall vs. spring 0.15 and 95% CI: 0.05 to 0.44, $P < 0.001$; OR winter vs. spring 0.13 and 95% CI: 0.03 to 0.57, $P = 0.007$; OR summer vs. spring 0.36 and 95% CI: 0.14 to 0.96, $P = 0.040$). Elevated serum NEFA ($P = 0.001$; OR 10.33; 95% CI: 2.53 to 42.13) and AST ($P = 0.035$; OR 1.03; 95% CI: 1.0 to 1.05) concentrations increased the risk of incurring hyperketonemia, whereas those of glucose ($P < 0.001$; OR 0.92; 95% CI: 0.89 to 0.96) reduced it. The 2 treatments containing hepato-protectors (CPP and SPP) were effective in preventing hyperketonemia (BHB ≥ 1.2 mmol/L) compared with CTR and MON, probably due to a more complete support to the metabolism as previously detailed. The risk of hyperketonemia was reduced by 35% and 73% in CPP compared with CTR and MON, respectively, and by 57% and 56% in SPP compared with CTR and MON, respectively. A greater risk (+4.5-fold) was evidenced at 14 compared with 7 d relative to calving (OR 4.46 and 95% CI: 1.26 to 15.8, $P = 0.020$). This result was in contrast to McArt et al. (2013) who reported the highest prevalence of hyperketonemia at 5 DIM. Moreover, spring was the season with the highest risk to develop hyperketonemia (+6.7-, +7.7-, and +2.8-fold relative to fall, winter, and summer). This result is in agreement with other studies and could be related to a different nutritional value of the provided feed or differences in milk yield (Suthar et al., 2013; Vanholder et al., 2015; Santschi et al., 2016).

Serum glucose was associated with a slight protective effect on hyperketonemia (-0.08%). Serum NEFA concentrations are considered a stronger indicator than those of BHB for metabolic dysfunctions (McArt et al.,

2013). However, serum NEFA and BHB concentrations are usually correlated, and consequently, it is difficult to separate their consequences (Rico and Barrientos-Blanco, 2024). However, herein, primiparous cows in the CTR group had a general hyperketonemic state without reaching an excessive lipomobilization; whereas both PPC and MPC in the MON group had an excessive lipomobilization without undergoing to hyperketonemic state. Nevertheless, serum NEFA concentration was the parameter that increased the most (+10.3-fold) the risk of hyperketonemia, followed by serum AST (+3%).

Milk Production and Reproductive Performance

Cows in MON and SPP had the greatest milk production compared with CTR and CPP cows considering the entire lactation period (43.5 and 42.9 kg/d for SPP and MON, respectively, vs. 42.0 and 41.7 kg/d for CTR and CPP; SEM = 0.49; $P < 0.05$). However, MON cows produced more milk than those in SPP between 25 and 100 DIM, whereas SPP cows produced similar amounts of milk as MON thereafter, with some periods (i.e., between 230 and 240) where milk production was greater in SPP than in MON at the later stage (Figure 4). The increased milking performance in MON cows during early lactation was probably related to an increase in propionate production at the ruminal level (Mammi et al., 2021). However, this potential change in propionic production in the rumen probably disappeared after 100 DIM (or before), and thus the positive consequences on milk yield also disappeared. Conversely, the improvement in milk yield in SPP cows, may have been related to a better AA metabolism during postpartum period with lower prevalence of hyperketonemia as discussed above.

All animals were AI between 80 and 90 DIM, with no differences between groups (82 DIM for CTR, 83 DIM for CPP, 83 DIM for SPP, and 86 DIM for MON; SEM = 1.40; $P = 0.32$). Pregnancy at first AI was greatest in

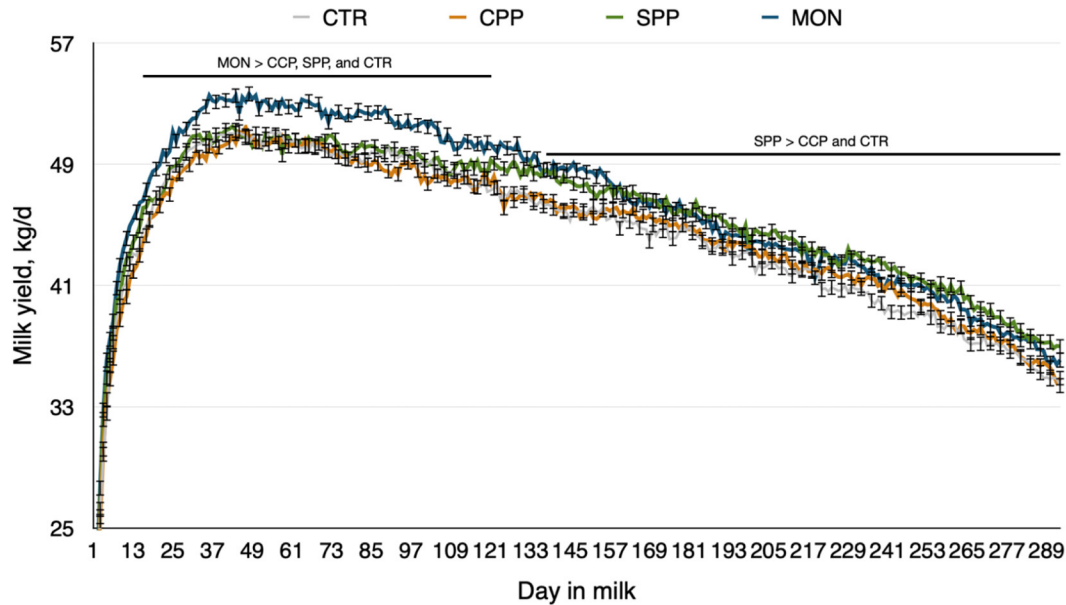


Figure 4. Milk yield as affected by treatments (CTR = no supplementation; CPP = cows received an i.m. injection of 2 mL of Bograss/10 kg of BW at 15 and 12 d before the expected calving date, and an i.m. injection of 20 mL of Erbacolina Plus at calving, and 2, 4, 6, 8, 10, and 12 d postpartum; SPP = cows received at 12 d before the expected calving and 6 d postpartum both drugs intramuscularly (2 mL/10 kg of BW and 70 mL/animal, respectively); MON = cows received one intraruminal bolus of monensin (32.4 g/animal) 21 d before the expected calving date. Error bars are SE.

SPP ($P = 0.04$) compared with all other groups (Table 3). Moreover, cows in SPP were the only ones that had greater odds of conceiving compared with CTR (OR = 1.16; 95% CI: 1.05 to 1.90). Similarly, cows in the SPP group had the greatest pregnancy per AI, followed by CTR and MON groups, and lowest in CPP ($P = 0.05$). After pregnancy confirmation at 68 d, pregnancy loss was near significance ($P = 0.09$) to be lowest in SPP cows (Table 3). It is possible that MON cows, that produced more milk early in lactation, had less nutrients available to divert to reproductive function, whereas the improved reproductive performance observed in SPP cows could be associated with a more constant and stable lactation curve over time and improved liver function and AA metabolism as discussed above.

Study Limitation

A possible limitation of this study is the omission of a placebo treatments in the CPP, SPP, and MON groups to standardize the number of administrations and animal handling performed. Differences in the number of occasions that animals were handled could have altered behavior and level of stress, thereby influencing the milk production and reproduction, especially in the CPP group, which received the highest number of interventions. Nevertheless, this field study aimed at evaluating the implementation of different hyperketonemia prevention protocols, as they would be performed on farms.

CONCLUSIONS

Supplementing hepato-protectors composed by AA, inositol, α -lipoic acid and cyanocobalamin reduced lipomobilization (lower NEFA) and metabolic stress (lower BHB) compared with control and monensin groups during the postpartum period, thereby promoting a better liver metabolism (indicated by lower serum AST and GGT concentrations). Furthermore, the administration of AA during postpartum (SPP) reduced muscle protein catabolism and subsequent urea production. Supplementing monensin improved milk production in early lactation, whereas supplementing hepato-protectors following a simplified protocol improved reproductive performance and improved milk yield at later stages of lactation. Therefore, the simplified protocol could represent a method for controlling postpartum hyperketonemia with limited animal handling.

NOTES

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F.C., and A.Ba.; Revising the original draft: A.V., A.Bi., A.Ba., B.C., and E.F. Animal care and procedures were conducted in accordance with the Guide for the Care and Use of Laboratory Animals and Directive 2010/63/EU for animal experiments (National law: D.L. 26/2014). The ethics statement was approved by the Animal Care and Use Committee of the University of Padua (protocol number 204359/2023). A. Valenza is an employee of Ceva Animal Health, Milan, Italy. The authors have not stated any other conflicts of interest.

Nonstandard abbreviations used: AI = artificial insemination; ALT = alanine amino-transferase; AST = amino-transferase; B = first hepato-protector containing AA (acetyl-Met, Ala, Arg, Thr, Gln), inositol, and cyanocobalamin; CK = creatinine kinase; CPP = complete preventive protocol; CTR = control; E = second hepato-protector containing acetyl-Met, α -lipoic acid, and cyanocobalamin; ER = endorumenal bolus of monensin; GGT = γ -glutamyl-transferase; LDH = lactate dehydrogenase; LFI = liver functionality index; MON = monensin; MPC = multiparous cows; NEB = negative energy balance; NEFA = nonesterified fatty acids; PPC = primiparous cows; SPP = simplified preventive protocol; TCA = tricarboxylic acid.

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